

Solutions for Problem Set 6

Mathematics for Social Scientists

EXERCISE 1. Let $\mathbf{u} = (1, 2)$, $\mathbf{v} = (0, 1)$, $\mathbf{w} = (1, -3)$, $\mathbf{x} = (1, 2, 0)$ and $\mathbf{z} = (0, 1, 1)$. Compute each expression (write “not defined” if sizes are incompatible):

- | | |
|---|---|
| (i) $\mathbf{u} + \mathbf{v}$ $(1, 3)$ | (vi) $\mathbf{u} + 2\mathbf{v}$ $(1, 4)$ |
| (ii) $-4\mathbf{w}$ $(-4, 12)$ | (vii) $\mathbf{u} - \mathbf{v}$ $(1, 1)$ |
| (iii) $\mathbf{x} + \mathbf{z}$ $(1, 3, 1)$ | (viii) $3\mathbf{x} + \mathbf{z}$ $(3, 7, 1)$ |
| (iv) $3\mathbf{z}$ $(0, 3, 3)$ | (ix) $-2\mathbf{x}$ $(-2, -4, 0)$ |
| (v) $2\mathbf{v}$ $(0, 2)$ | (x) $\mathbf{w} + 2\mathbf{x}$ not defined |

EXERCISE 2. Find the length (Euclidean norm) of each vector and (optionally) sketch:

- (a) $(3, 4)$ (b) $(0, -3)$ (c) $(1, 1, 1)$ (d) $(3, 3)$ (e) $(-1, -1)$ (f) $(1, 2, 3)$ (g) $(2, 0)$.

(a) 5; (b) 3; (c) $\sqrt{3}$; (d) $\sqrt{18}$; (e) $\sqrt{2}$; (f) $\sqrt{14}$; (g) 2.

EXERCISE 3. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$. If possible, compute $A + B$ and $A \times B$ for the following matrices B .

(i) $B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

$$A + B = \begin{pmatrix} 3 & 2 \\ 3 & 2 \end{pmatrix}, A \times B = \begin{pmatrix} 2 & 4 \\ 6 & 0 \end{pmatrix}$$

(ii) $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$A + B = \begin{pmatrix} 1 & 3 \\ 4 & 0 \end{pmatrix}, A \times B = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

(iii) $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

$$A + B \text{ is not defined. } A \times B = \begin{pmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}$$

(iv) $B = A^T$

$$A + B \begin{pmatrix} 2 & 5 \\ 5 & 0 \end{pmatrix}, A \times B = \begin{pmatrix} 5 & 3 \\ 3 & 9 \end{pmatrix}$$

EXERCISE 4. For each of the following matrices A , define a vector $v \neq 0$ such that $A \cdot v = v$. If it doesn't exist, state why.

$$(i) \quad A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

No such vector exists.

$$(ii) \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is one vector that satisfies $A \cdot v = v$

$$(iii) \quad A = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 1 & -1 & 0 \end{pmatrix}$$

$v = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is one vector that satisfies $A \cdot v = v$

EXERCISE 5. Let $A, B \in \mathbb{R}^{3 \times 3}$ be diagonal matrices. Show that $A \times B$ is diagonal.

$$\begin{pmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{pmatrix} \times \begin{pmatrix} b_{11} & 0 & 0 \\ 0 & b_{22} & 0 \\ 0 & 0 & b_{33} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} & 0 & 0 \\ 0 & a_{22}b_{22} & 0 \\ 0 & 0 & a_{33}b_{33} \end{pmatrix}$$

EXERCISE 6. Let

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & -1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & -1 \\ 4 & -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad E = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Compute each matrix if it is defined:

$$\begin{aligned}
 \text{(i)} \quad A + B &= \begin{pmatrix} 2 & 4 & 0 \\ 4 & -2 & 4 \end{pmatrix} \\
 \text{(ii)} \quad A - D &= \text{undefined} \\
 \text{(iii)} \quad 3B &= \begin{pmatrix} 0 & 3 & -3 \\ 12 & -3 & 6 \end{pmatrix} \\
 \text{(iv)} \quad DC &= \begin{pmatrix} 5 & 3 \\ 4 & 1 \end{pmatrix} \\
 \text{(v)} \quad B^T &= \begin{pmatrix} 0 & 4 \\ 1 & -1 \\ -1 & 2 \end{pmatrix} \\
 \text{(vi)} \quad A^T C^T &= \begin{pmatrix} 2 & 6 \\ 1 & 10 \\ 5 & 1 \end{pmatrix} \\
 \text{(vii)} \quad C + D &= \begin{pmatrix} 3 & 3 \\ 4 & 0 \end{pmatrix} \\
 \text{(viii)} \quad B - A &= \begin{pmatrix} -2 & -2 & -2 \\ 4 & 0 & 0 \end{pmatrix} \\
 \text{(ix)} \quad AB &= \text{undefined}
 \end{aligned}$$

$$\begin{aligned}
 \text{(x)} \quad CE &= \begin{pmatrix} -1 \\ 4 \end{pmatrix} \\
 \text{(xi)} \quad -D &= \begin{pmatrix} -2 & -1 \\ -1 & -1 \end{pmatrix} \\
 \text{(xii)} \quad (CE)^T &= \begin{pmatrix} -1 & 4 \end{pmatrix} \\
 \text{(xiii)} \quad B + C &= \text{undefined} \\
 \text{(xiv)} \quad D - C &= \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix} \\
 \text{(xv)} \quad CA &= \begin{pmatrix} 2 & 1 & 5 \\ 6 & 10 & 1 \end{pmatrix} \\
 \text{(xvi)} \quad EC &= \text{undefined} \\
 \text{(xvii)} \quad (CA)^T &= \begin{pmatrix} 2 & 6 \\ 1 & 10 \\ 5 & 1 \end{pmatrix} \\
 \text{(xviii)} \quad E^T C^T &= \begin{pmatrix} -1 & 4 \end{pmatrix}
 \end{aligned}$$